NETWORK COMPONENTS,TRANSMISSION MEDIUM & TOPOLOGY NOTES

1. **Network packet**

   A network packet is a formatted unit of data (a list of bits or bytes, usually a few tens of bytes to a few kilobytes long) carried by a packet-switched network.

2. ***Repeaters***

   A repeater is an electronic device that receives a network signal, cleans it of unnecessary noise and regenerates it. The signal is retransmitted at a higher power level, or to the other side of an obstruction, so that the signal can cover longer distances without degradation

3. **Hubs**

   Is a device for connecting multiple Ethernet devices together and making them act as a single network segment. It has multiple input/output (I/O) ports, in which a signal introduced at the input of any port appears at the output of every port except the original incoming

4. ***Bridges***

5. A network bridge connects and filters traffic between two network segments at the data link layer (layer 2) of the OSI model to form a single network. This breaks the network's collision domain **Network Switch**

   Is a computer that connects devices together on a computer network, by using packet switching to receive, process and forward data to the destination device

6. ***Routers***

   A router is an internetworking device that forwards packets between networks by processing the routing information included in the data packet.

7. ***Modems***

   Modems (MOdulator-DEModulator) are used to connect network nodes via wire not originally designed for digital network traffic, or for wireless. To do this one or more carrier signals are modulated by the digital signal to produce an analog signal that can be tailored to give the required properties for transmission. Modems are commonly used for telephone lines, using a Digital Subscriber Line technology.

8. ***Firewalls***

   A firewall is a network device for controlling network security and access rules. Firewalls are typically configured to reject access requests from unrecognized sources while allowing actions

from recognized ones. The vital role firewalls play in network security grows in parallel with the constant increase in cyber-attacks.

Transmission Medium

Various transmission media can be used for transfer of data. These transmission media may be of two types −

- **Guided** − In guided media, transmitted data travels through cabling system that has a fixed path. For example, copper wires, fibre optic wires, etc.
- **Unguided** − In unguided media, transmitted data travels through free space in form of electromagnetic signal. For example, radio waves, lasers, etc.

Each transmission media has its own advantages and disadvantages in terms of bandwidth, speed, delay, cost per bit, ease of installation and maintenance, etc.

Guided Transmission Medium

1. Twisted Pair Cable

Copper wires are the most common wires used for transmitting signals because of good performance at low costs. They are most commonly used in telephone lines. However, if two or more wires are lying together, they can interfere with each other's signals. To reduce this electromagnetic interference, pair of copper wires are twisted together in helical shape like a DNA molecule. Such twisted copper wires are called **twisted pair**. To reduce interference between nearby twisted pairs, the twist rates are different for each pair.



Up to 25 twisted pair are put together in a protective covering to form twisted pair cables that are the backbone of telephone systems and Ethernet networks.

Advantages of twisted pair cable

Twisted pair cable are the oldest and most popular cables all over the world. This is due to the many advantages that they offer −

- Trained personnel easily available due to shallow learning curve
- Can be used for both analog and digital transmissions
- Least expensive for short distances
- Entire network does not go down if a part of network is damaged

Disadvantages of twisted pair cable

With its many advantages, twisted pair cables offer some disadvantages too −

- Signal cannot travel long distances without repeaters
- High error rate for distances greater than 100m

- Very thin and hence breaks easily
- Not suitable for broadband connections

Shielding twisted pair cable

To counter the tendency of twisted pair cables to pick up noise signals, wires are shielded in the following three ways −

- Each twisted pair is shielded.
- Set of multiple twisted pairs in the cable is shielded.
- Each twisted pair and then all the pairs are shielded.

Such twisted pairs are called **shielded twisted pair (STP) cables**. The wires that are not shielded but simply bundled together in a protective sheath are called **unshielded twisted pair (UTP) cables**. These cables can have maximum length of 100 metres.

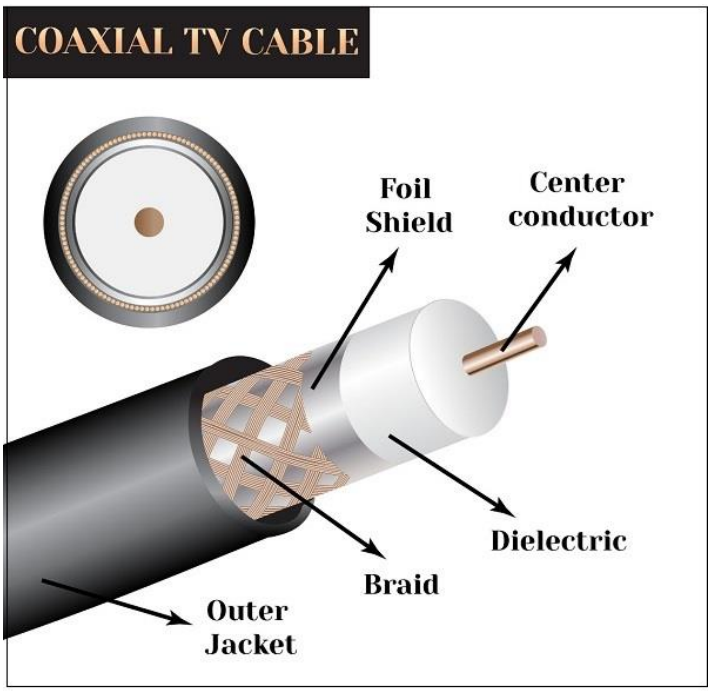
Shielding makes the cable bulky, so UTP are more popular than STP. UTP cables are used as the last mile network connection in homes and offices.

2. Coaxial Cable

Coaxial cables are copper cables with better **shielding** than twisted pair cables, so that transmitted signals may travel longer distances at higher speeds. A coaxial cable consists of these layers, starting from the innermost −

- Stiff copper wire as **core**
- **Insulating material** surrounding the core
- Closely woven braided mesh of **conducting material** surrounding the **insulator**
- Protective **plastic sheath** encasing the wire

Coaxial cables are widely used for **cable TV** connections and **LANs**.



Advantages of Coaxial Cables

These are the advantages of coaxial cables −

- Excellent noise immunity
- Signals can travel longer distances at higher speeds, e.g. 1 to 2 Gbps for 1 Km cable
- Can be used for both analog and digital signals
- Inexpensive as compared to fibre optic cables
- Easy to install and maintain

Disadvantages of Coaxial Cables

These are some of the disadvantages of coaxial cables −

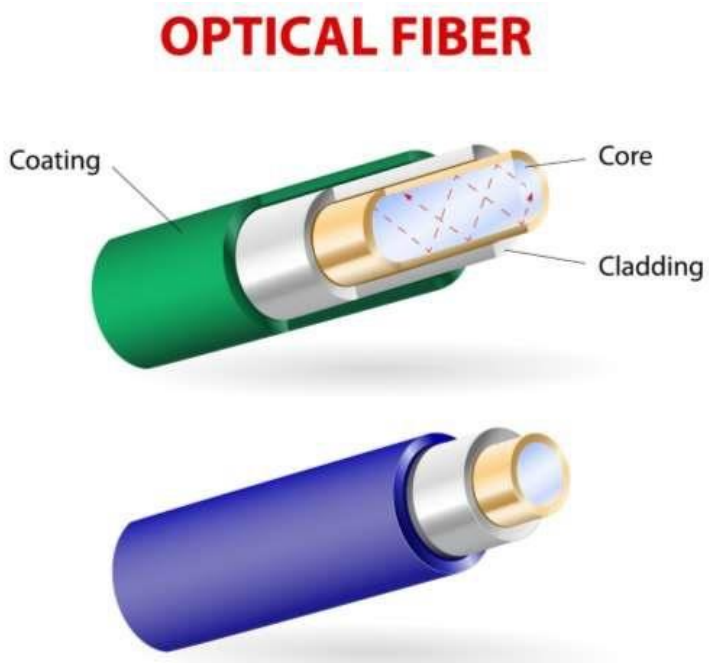
- Expensive as compared to twisted pair cables
- Not compatible with twisted pair cables

3. Optical Fibre

Thin glass or plastic threads used to transmit data using light waves are called **optical fibre**. Light Emitting Diodes (LEDs) or Laser Diodes (LDs) emit light waves at the **source**, which is read by a **detector** at the other end. **Optical fibre cable** has a bundle of such threads or fibres bundled together in a protective covering. Each fibre is made up of these three layers, starting with the innermost layer −

- **Core** made of high quality **silica glass** or **plastic**
- **Cladding** made of high quality **silica glass** or **plastic**, with a lower refractive index than the core
- Protective outer covering called **buffer**

Note that both core and cladding are made of similar material. However, as **refractive index** of the cladding is lower, any stray light wave trying to escape the core is reflected back due to **total internal reflection**.



Optical fibre is rapidly replacing copper wires in telephone lines, internet communication and even cable TV connections because transmitted data can travel very long distances without weakening. **Single node** fibre optic cable can have maximum segment length of 2 kms and bandwidth of up to 100 Mbps. **Multi-node** fibre optic cable can have maximum segment length of 100 kms and bandwidth up to 2 Gbps.

Advantages of Optical Fibre

Optical fibre is fast replacing copper wires because of these advantages that it offers −

- High bandwidth
- Immune to electromagnetic interference
- Suitable for industrial and noisy areas
- Signals carrying data can travel long distances without weakening

Disadvantages of Optical Fibre

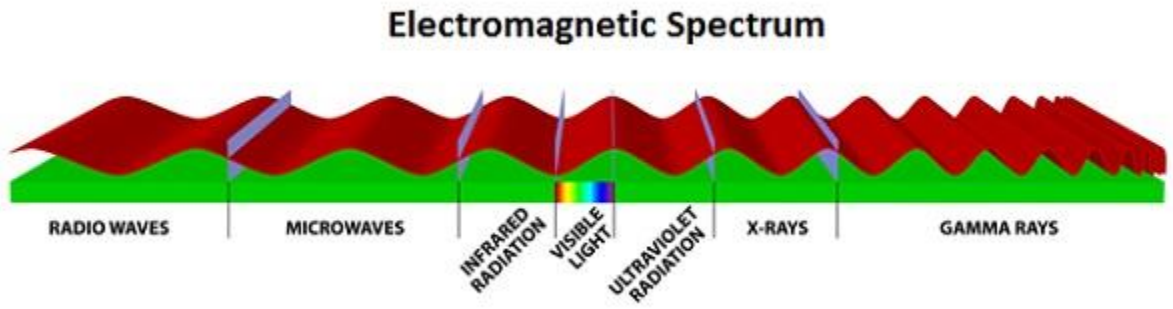
Despite long segment lengths and high bandwidth, using optical fibre may not be a viable option for every one due to these disadvantages −

- Optical fibre cables are expensive
- Sophisticated technology required for manufacturing, installing and maintaining optical fibre cables
- Light waves are unidirectional, so two frequencies are required for full duplex transmission

Unguided Medium Transmission Medium

1. Infrared

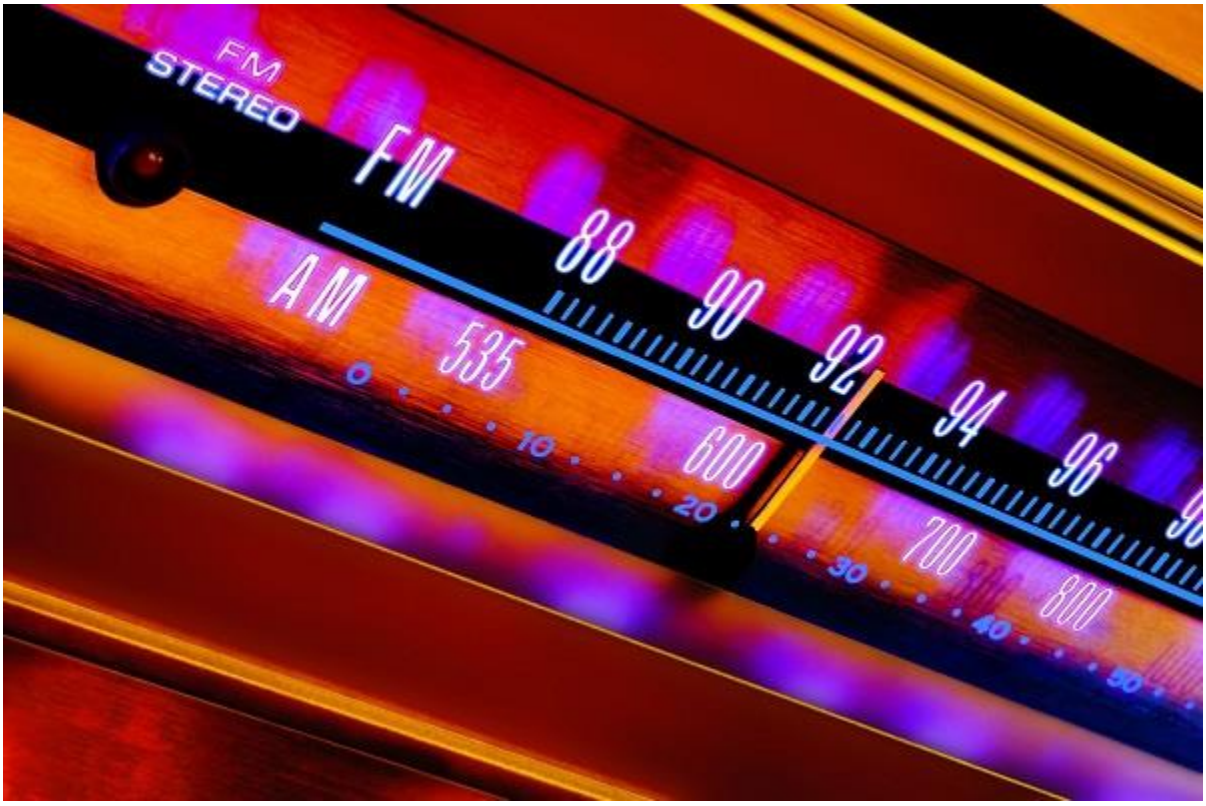
Low frequency infrared waves are used for very short distance communication like TV remote, wireless speakers, automatic doors, hand held devices etc. Infrared signals can propagate within a room but cannot penetrate walls. However, due to such short range, it is considered to be one of the most secure transmission modes.



2. Radio Wave

Transmission of data using radio frequencies is called **radio-wave transmission**. We all are familiar with radio channels that broadcast entertainment programs. Radio stations transmit radio waves using **transmitters**, which are received by the receiver installed in our devices.

Both transmitters and receivers use antennas to radiate or capture radio signals. These radio frequencies can also be used for **direct voice communication** within the **allocated range**. This range is usually 10 miles.



Advantages of Radio Wave

These are some of the advantages of radio wave transmissions −

- Inexpensive mode of information exchange
- No land needs to be acquired for laying cables
- Installation and maintenance of devices is cheap

Disadvantages of Radio Wave

These are some of the disadvantages of radio wave transmissions −

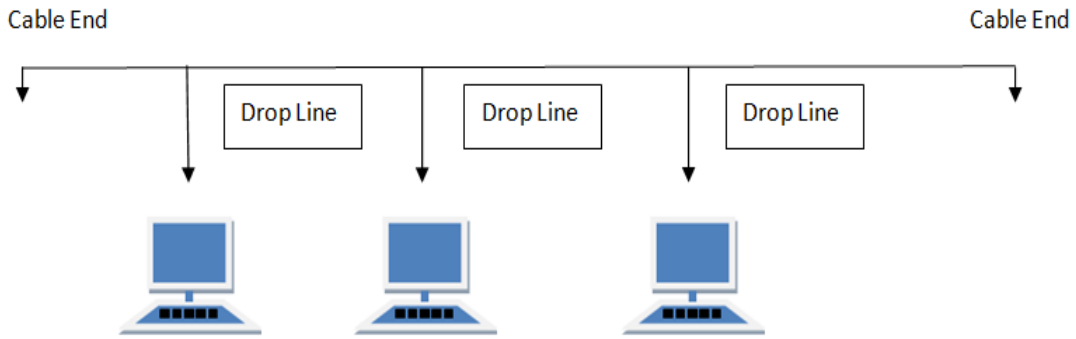
- Insecure communication medium
- Prone to weather changes like rain, thunderstorms, etc

TYPES OF NETWORK TOPOLOGY

Network Topology is the schematic description of a network arrangement, connecting various nodes (sender and receiver) through lines of connection.

BUS Topology

Bus topology is a network type in which every computer and network device is connected to single



cable. When it has exactly two endpoints, then it is called **Linear Bus topology**.

Features of Bus Topology

1) It transmits data only in one direction.
2) Every device is connected to a single cable

Advantages of Bus Topology

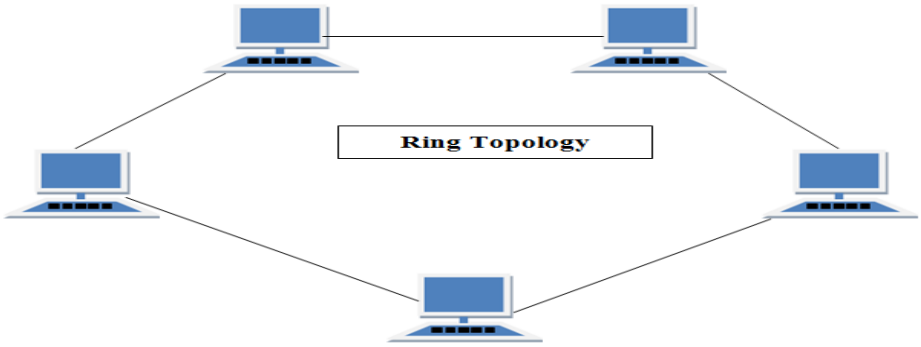
1) It is cost effective.
2) Cable required is least compared to other network topology.
3) It is easy to understand.
4) Easy to expand joining two cables together.
5) It works well when you have a small network.
6) It's the easiest network topology to implement.
7) It requires less cable length than a star topology.

Disadvantages of Bus Topology

1) Cables fails then whole network fails.
2) If network traffic is heavy or nodes are more the performance of the network decreases.
3) Cable has a limited length.
4) It is slower than the ring topology.
5) It can be difficult to identify the problems if the whole **network** goes down.
6) It can be hard to troubleshoot individual device issues.
7) **Bus topology** is not great for large networks.
8) Terminators are required for both ends of the main cable.
9) Additional devices slow the **network** down.

RING Topology

It is called ring topology because it forms a ring as each computer is connected to another computer, with the last one connected to the first. Exactly two neighbors for each device.



Features of Ring Topology

1) A number of repeaters are used for Ring topology with large number of nodes, because if someone wants to send some data to the last node in the ring topology with 100 nodes, then the data will have to pass through 99 nodes to reach the 100th node. Hence to prevent data loss repeaters are used in the network.
2) The transmission is unidirectional, but it can be made bidirectional by having 2 connections between each Network Node, it is called **Dual Ring Topology**.
3) In Dual Ring Topology, two ring networks are formed, and data flow is in opposite direction in them. Also, if one ring fails, the second ring can act as a backup, to keep the network up.
4) Data is transferred in a sequential manner that is bit by bit. Data transmitted, has to pass through each node of the network, till the destination node.

Advantages of Ring Topology

1) Transmitting network is not affected by high traffic or by adding more nodes, as only the nodes having tokens can transmit data.
2) Cheap to install and expand

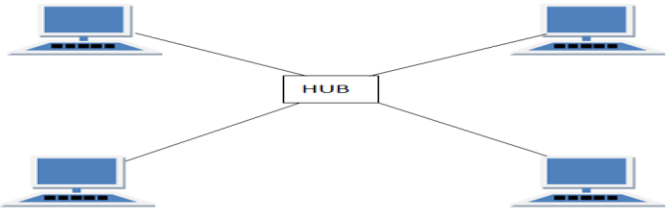
Disadvantages of Ring Topology

1. Troubleshooting is difficult in ring topology.
2. Adding or deleting the computers disturbs the network activity.

1. Failure of one computer disturbs the whole network.

STAR Topology

In this type of topology all the computers are connected to a single hub through a cable. This hub



is the central node and all others nodes are connected to the central node.

Features of Star Topology

1. Every node has its own dedicated connection to the hub.
2. Hub acts as a repeater for data flow.
3. Can be used with twisted pair, Optical Fibre or coaxial cable.

Advantages of Star Topology

1) Fast performance with few nodes and low network traffic.
2) Hub can be upgraded easily.
3) Easy to troubleshoot.
4) Easy to setup and modify.
5) Only that node is affected which has failed, rest of the nodes can work smoothly.

Disadvantages of Star Topology

1) Cost of installation is high.
2) Expensive to use.
3) If the hub fails then the whole network is stopped because all the nodes depend on the hub.
4) Performance is based on the hub that is it depends on its capacity

MESH Topology

It is a point-to-point connection to other nodes or devices. All the network nodes are connected to each other. Mesh has n(n-1)/2 physical channels to link n devices.

There are two techniques to transmit data over the Mesh topology, they are :

1) Routing
2) Flooding

Routing

In routing, the nodes have a routing logic, as per the network requirements. Like routing logic to direct the data to reach the destination using the shortest distance. Or, routing logic which has

information about the broken links, and it avoids those node etc. We can even have routing logic, to re-configure the failed nodes.

Flooding

In flooding, the same data is transmitted to all the network nodes; hence no routing logic is required. The network is robust, and it's very unlikely to lose the data. But it leads to unwanted load over the network.



Types of Mesh Topology

1) **Partial Mesh Topology:** In this topology some of the systems are connected in the same fashion as mesh topology but some devices are only connected to two or three devices.
2) **Full Mesh Topology:** Each and every nodes or devices are connected to each other.

Features of Mesh Topology

1. Fully connected.
2. Robust.
3. Not flexible.

Advantages of Mesh Topology

1. Each connection can carry its own data load.
2. It is robust.
3. Fault is diagnosed easily.
4. Provides security and privacy.

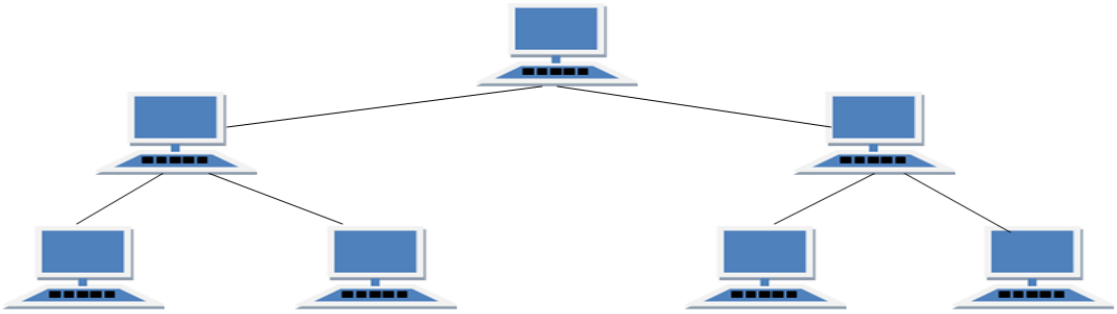
Disadvantages of Mesh Topology

1. Installation and configuration is difficult.

2. Cabling cost is more.
3. Bulk wiring is required.

TREE Topology

It has a root node and all other nodes are connected to it forming a hierarchy. It is also called



hierarchical topology. It should at least have three levels to the hierarchy.

Features of Tree Topology

1. Ideal if workstations are located in groups.
2. Used in Wide Area Network.

Advantages of Tree Topology

1. Extension of bus and star topologies.
2. Expansion of nodes is possible and easy.
3. Easily managed and maintained.
4. Error detection is easily done.

Disadvantages of Tree Topology

1. Heavily cabled.
2. Costly.
3. If more nodes are added maintenance is difficult.
4. Central hub fails, network fails.

HYBRID Topology

It is two different types of topologies which is a mixture of two or more topologies. For example if in an office in one department ring topology is used and in another star topology is used, connecting these topologies will result in Hybrid Topology (ring topology and star topology).

Features of Hybrid Topology

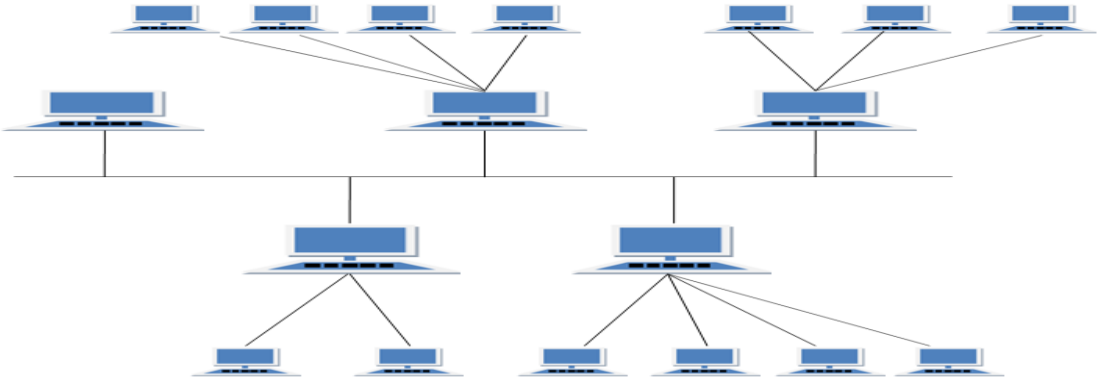
1. It is a combination of two or more topologies
2. Inherits the advantages and disadvantages of the topologies included

Advantages of Hybrid Topology

1. Reliable as Error detecting and trouble shooting is easy.
2. Effective.
3. Scalable as size can be increased easily.
4. Flexible.

Disadvantages of Hybrid Topology

1. Complex in design.



2. Costly.

ADVANTAGES OF COMPUTER NETWORKING

1. **Allows File Sharing**

   Computer networking allows accessibility for the people to share their files easily. This greatly helps the people in saving their time and effort since they could already share some of the important files they need to share with other people effectively and accordingly.

2. **Provides Convenient Resource Sharing**

   This is very important particularly for those larger companies who are greatly in need to produce huge number of resources to be shared to all the people. Since this is a computer based works, you will be assured that the resources you wanted to give to the people would be completely shared if you are going to connect them in a network which you are also using.

3. **Inexpensive System**

   Installing computer networking software in your gadget would not cost you too much since you are assured that the software is durable and can effectively share information to the people you need it to. You need not to change the software regularly since it is very simple and can last for a couple of years.

4. **Flexible to be Used**

   Computer networking is known to be very flexible since you are given the chance to explore everything about the software without affecting its functionality. You will have accessibility to all the information that needs to be shared.

5. **Increase in Storage Capacity of the Software**

   Since you are going to share files and resources to other people, you need to make sure that all the data and files are properly stored in the system. With computer networking you are assured that all the files would be stored properly depending on your saved file.

DISADVANTAGES OF COMPUTER NETWORKING

1. **Security Difficulties**

   Since there are already huge number of people who are using computer networking in sharing some of their files and resources, your security would be always at risk. There might be illegal activities that will occur that you need to be aware and be careful all the time.

2. **Presence of Computer Viruses and Other Malwares**

   There are instances that the stored file you have in your gadget is already destroyed because there are already viruses that corrupt your file. Hence you need to have regular check-up on your gadget at the same time to the stored files you have.

3. **Lack of Independence**

   Since computer networking is a process operated through computers, people are already relying more of the works of the computer rather than exerting an effort for their works.

   Computer networking will always give you a fast and convenient way to share and transfer information to the people. However, people need to make sure that they are always aware of the consequences that this particular system would be giving them. They should not always rely on this system even if there are several advantages that this is bringing to prevent the presence of flaws and other malfunctions of the system.